ANNEX A

ASR System Hardware and Software

The hardware consists of a desktop personal computer (AST Advantage! Pro with a 80486DX 33MHz processor, 430 MByte hard drive and 8MByte RAM), a sound card (Gravis Ultrasound Max capable of 16 bit 48 KHz stereo recording and playback), a high quality microphone (Realistic Super Cardiod Dynamic Microphone with 80 to 15000 Hz frequency response), a stereo tape deck (Realistic SCT-90 with metal bias recording option), a metal bias cassette tape (Sony Digital Excellence Metal SR type IV metal bias 70microsec EQ), and a home made microphone enclosure consisting of 3 cubic foot cardboard box lined with egg cartons.

The software used runs within MicroSoft's Windows For Workgroups version 3.11 running on top of MicroSoft's DOS version 6.2. Mathwork's Matlab for Windows version 4.0 was used to perform the speech signal feature extraction and simulation of the fuzzy ART Map neural network. Automated speech recording is performed using GoldWave version 2.11 controlled by a Windows script file program called Stdemo Player version 1.5. The compression of digitized speech files is accomplished using WazWave version 0.1. The ARJ compression program was used to make a backup achive copy of the database. The FORM1968 program was used to reformat floppy disks to a 1968KByte capacity.